



Programming with Python

Subject: Programming with Python (042024-MAL)

Intakes: UCDF2307/ICT(SE)

Group: Group 28

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1.0 Introduction

The Ministry of Health requires an efficient system to manage the inventory of personal protective equipment (PPE) received from multiple vendors and distributed to multiple hospitals. This system is highly needed as it ensures that healthcare facilities have adequate equipment to protect healthcare workers and patients, especially during times of increased demand. This project aims to develop a Python-based inventory management system that tracks the receipt, storage, and distribution of various PPE items, ensuring accurate stock levels and timely supply to hospitals.

The system will include features such as secure login capabilities for inventory managers, initial inventory setup, and the ability to update stock levels when PPE items are received from vendors or distributed to hospitals. By implementing this system, the Ministry of Health aims to streamline its inventory management processes, reduce errors, and maintain a steady supply of essential PPE items.

2.0 Assumptions

To effectively design and implement the inventory management system, several assumptions have been made.

1. “ppe.txt” will be created only once when the program is executed for the first time. Both “supplier.txt” and “ppe.txt” must be created before accessing the home page.
2. Files that have already been created cannot be created again; only rewriting or updating them is allowed.
3. All information inserted into the txt files must match the format of the expected input exactly.
4. Each item is supplied by exactly one supplier, meaning that a particular item cannot be supplied by more than one supplier. However, one supplier can supply more than one type of item.
5. There can be only maximum four inventory controllers. If a fifth user attempts to sign up, the registration will be denied.
6. Inventory controllers can attempt to log in a maximum of three times. If all three attempts fail, the account will be permanently terminated and removed from the system. Once an account is removed, there will be three remaining inventory controllers, allowing a new user to sign up and access the system.
7. The initial quantity of each item is 100 when “ppe.txt” is created.
8. All ppe items are measured by boxes.
9. There is minimum 3 hospitals in the state. As only 3 or 4 hospitals can exist.
10. Only 3 or 4 suppliers can exist.
11. Supplier details can be updated in “supplier.txt”, hospital details could be updated in “hospital.txt”.
12. All distributions are recorded in the “distribution.txt”.
13. The database will be refreshed every year, which means that a month will not repeat twice.

3.0 Assignment Workload Matrix

Part	Component	Student 1 Name: Elvan Sea Meng Jin	Student 2 Name: Lai Xiao Chun	Student 3 Name: Kee Genni	Student 4 Name: Jayden Foo Lok Hin	Total
1	a) Introduction & Assumptions	20%	30%	30%	20%	100%
1	b) Pseudocode	30%	20%	30%	20%	100%
1	c) Flowchart	20%	20%	30%	30%	100%

4.0 Design of the program

4.1 Pseudocode

#Defining Login Function

START

Function login (users, attempts, max_attempts=3):

 Loop until True:

 Prompt user for username

 #If username is not valid

 While username not in users:

 Print("This username is not found. Please try again")

 Prompt user for username

 End While

 Prompt user for password

 #Checking if user login attempt is more than 3

 If username is in (attempts) dictionary and attempts[username] >= max_attempts:

 Print("Too many failed attempts. Access terminated.")

 Return False #This will quit the program

 End If

 If (username and password) in users dictionary: #Check dictionary

 Print("Login successful")

 //Redirect to main page//

 Return True #Quit function and redirect to home function

 Else:

 #Check if username exists in attempts (login before but failed)

 If username is in attempts:

 Attempts[username] +=1

 Else:

 Attempts[username] = 1

 End If

 If (attempts) dictionary username more than max_attempts:

 Print("You are banned due to too many attempts")

 Remove username and password from users dictionary

 Set user count -=1

 Return False

 //Redirect to home page//

 Else:

 Print("Login failed, please try again.")

 End If

 End If

 End Loop

End Function

END

#Defining Sign Up Function

START

Function sign_up (users, user_count):

 Loop until True:

 #Check if the user_count reached the maximum number

 If user_count more than or equal to 4:

 Print("The maximum number of registered users has been reached.")

 Return user_count #Quit sign_up function

 Else:

 Prompt user for username

 If username in users dictionary:

 Print("Username exists. Please choose a different username.")

 Else:

 Prompt user for password

 Add username and password in users dictionary

 Print ("Sign up successful.")

 Set User_count +=1

 Print ("You have logged in. Welcome to inventory management system")

 //Redirect to main page//

 Return user_count #return latest user_count

 End If

 End If

 End Loop

End Function

END

#Defining Function to Create ppe.txt

START

Function ppe_file (ppe_path):

 #Check if the file exists or not

 If ppe_path does not exist:

 Print (ppe_path, “does not exist. Creating a new file”)

 #Open the file in write mode

 Open file at ppe_path in write mode

 #Loop until prompt from user

 Loop until True:

 Prompt user (“Enter text to write into this file or ‘done’ to finish”)

 If user inputs ‘done’:

 Break

 Else:

 Insert user input into ppe_path

 #expected input

 #(Item Code, Item Name, Quantity(Box), Supplier Code, Current Date)

 #(HC, Head Cover, 100, S01, January 27, 2024)...

 End If

 End Loop

 Close ppe_path

 Print (ppe_path, “has been created with your input.”)

Else:

 Pass

End If

End Function

END

#Defining Function to Create supplier.txt

START

Function supplier_file(supplier_path)

 If supplier_path does not exists:

 Print (supplier_path, "does not exist, creating a new file")

 Set supplier_path as a file and set to write mode

 Set variable supply_count = 0

 Loop until True:

 Prompt user ("Please enter supplier details or type 'done' to finish")

 If user input == "done" or supply_count >=4:

 Break

 Else:

 Insert supplier_code, supplier_name, supplier_address, supplier_contact into supplier_path file

 #expected input

 #(Supplier Code, Supplier Name, Supplier Address, Supplier Contact)

 Set supply_count += 1

 End If

End Loop

Close supplier_path

Print (supplier_path, "has been created with your input.")

Else

 Pass

End if

End Function

END

#Defining Function to Create hospital.txt

START

Function hospital_file(hospital_path)

If hospital.txt does not exists:

Loop until True:

Prompt user (“Would you like to create hospital.txt file?, ‘1’ for Yes, ‘2’ for No”)

If user inputs 1:

Set hospital_path as a file and set to write mode

Set variable hospital_count = 0

Loop until True:

Prompt user (“Please enter hospital details or type ‘done’ to finish”)

If user input == “done” or hospital_count >= 4:

Break

Else:

Insert hospital_code, hospital_name into hospital_path

Set hospital_count += 1

End If

End Loop

Else if user inputs 2:

Break #Send back to home page

Else:

Print(“Please enter the correct input”)

End If

End Loop

Close hospital_path

Else:

Print (“hospital.txt already exists. Returning to home page.”)

Break #Return to home page

End if

End Function

END

```
#Defining Function to modify distribution.txt
```

```
START
```

```
Function distribution_file(distribution_path)
```

```
    Loop until True:
```

```
        Prompt user "What item do you want to distribute? (please type in Item Code)"
```

```
        #Check if the item exists
```

```
        If user input not found in ppe_path:
```

```
            Print ("This item does not exist. Please enter a valid code")
```

```
        Else:
```

```
            Prompt user ("Please input the amount of items to be distributed")
```

```
        Try:
```

```
            Set user input to int
```

```
            #Check if item quantity is enough to distribute
```

```
            If user input > ppe_path(quantity) #By Item Code:
```

```
                Print ("This item is not enough for distribution.")
```

```
            Else:
```

```
                Prompt user ("Which hospital do you want to distribute to? (please type in Hospital Code )")
```

```
                #Check if this hospital exists
```

```
                If user input not found in hospital_path:
```

```
                    Print ("This hospital does not exist. Please enter a valid code")
```

```
                Else:
```

```
                    Set distribution.txt file as read+ mode
```

```
                    Set ppe_path file as read + mode
```

```
                    Set ppe.txt quantity -= user input quantity #By Item Code
```

```
                    Insert Item Code, Quantity, Hospital Code, and Current Date into distribution_path
```

```
                    Print("Your distribution has been recorded successfully.")
```

```
                    Break
```

```
            End If
```

```
        End If
```

```
    Except ValueError:
```

```
        Print("This is not a valid number")
```

```
    End If
```

```
End Loop
```

```
Close ppe_path file, distribution_path file
```

```
End Function
```

```
END
```

#Defining Function to sum up quantity distribution by the same hospital

START

Function distribution_hospital()

Open hospital_path file in read mode

Hospitals = {} #Create Dictionary

For each line in hospital_path:

Split line into parts by “,”

Extract and Insert Hospital_code, Hospital_name into Hospitals Dictionary

Close hospital_path file

End For Loop

Open distribution_path file in read mode

Distribution = {} #Create Dictionary

For each line in distribution path:

Split line into parts by “,”

Extract and Insert Item_Code, Quantity, Hospital_Code into Distribution Dictionary

#Sum up Quantity for same hospital

If hospital_code in distribution:

distribution[hospital_code] += quantity

Else:

distribution[hospital_code] = quantity

End If

End For Loop

#Print list of hospitals with quantities of distributed items

For hospital_code in hospitals:

Hospital_name = hospitals[hospital_code]

Total_quantity = distribution.get(hospital_code, 0) #return 0 if no items distributed to that hospital

Print(Item_Code, “has been distributed”, quantity, “to”, hospital_code)

End For Loop

Break

End Function

END

#Defining Function for Home Page

START

Function home():

```

    #after user successfully login
    Set ppe_path = "ppe.txt"
    Set supplier_path = "supplier.txt"
    Set hospital_path = "hospital.txt"
    Set distribution_path = "distribution.txt"
    Print("Welcome to the menu!")

    Call Function ppe_file(ppe_path) #ppe.txt
    Call Function supplier_file(supplier_path) #supplier.txt

    Loop until True:
        Prompt user ("What would you like to do?
        \nPrompt '1' to Create Hospital File
        \nPrompt '2' to Update Inventory
        \nPrompt '3' to Update Supplier Details
        \nPrompt '4' to Update Hospital Details
        \nPrompt '5' to Track Inventory Items
        \nPrompt '6' to Search Distribution List
        \nPrompt '7' to Print Report
        \nPrompt '8' to Log Out")

        If user inputs 1: #Create Hospital File
            Call Function hospital_file(hospital_path)

        Else if user inputs 2: #Update Inventory
            Loop until True:
                Prompt user ("Would you like to add to the inventory or distribute inventory? [1 to Add Inventory, 2 to Distribute Inventory]")
                If user inputs 1:
                    Loop until True:
                        Prompt user ("What item do you want to add? (please type in Item Code)")
                        #Check if the item exists
                        If user input not found in ppe_path:
                            Print ("This item does not exist. Please enter a valid code")
                        Else:
                            Set ppe_path file to read + mode
                            Prompt user ("How many items?")
                            Try:
                                Set user input to int
                                Set ppe_path file (quantity) += user input #By Item Code
                                Set ppe_path file date to Current Date
                                Close ppe_path
                                Break
                            Except ValueError:
                                Print ("This is not a valid number")
                        End If
                    End Loop
                End If

            Else if user inputs 2:
                If hospital_path exists:
                    If distribution_path exists:
                        Call Function distribution_file(distribution_path)
                        Break
                    Else:
                        Set distribution_path as a file
                        Call Function distribution_file(distribution_path)
                        Break
                    End If
                Else:
                    Print("Please create hospital file before distribute inventory")
                    Call Function hospital_file(hospital_path)

            Else:
                Print ("This is not a valid answer. Please try again")

            End If

        End Loop

        Else if user inputs 3: #Update Supplier Details
            Loop until True:
                Prompt user ("Please enter the supplier code")

                #Check if supplier code exists
                If user input is not found in supplier_path:
                    Print ("This code does not exist, please try again")

            Else:
                Loop until True:
                    Prompt user ("Which detail do you want to edit? [Name, Address, Contact])
                    Set supplier_path to read +
                    If user input == "Name"

```

```

        Prompt user ("Please enter the new name")
        Edit supplier to new supplier name #By Supplier Code
        Break

    Else If user input == "Address":
        Prompt user ("Please enter the new address")
        Edit supplier address to new supplier address #By Supplier Code
        Break

    Else If user input == "Contact":
        Prompt user ("Please enter the new contact")
        Edit supplier contact to new supplier contact #By Supplier Code
        Break

    Else:
        Print ("Please enter the correct details.")

    End If

End Loop

End If

End Loop

Else if user inputs 4: #Update Hospital Details

    #Check if hospital.txt exist

    If hospital_path exists:
        Loop until True:
            Prompt user ("Please enter the hospital code")

            #Check if hospital code exists
            If user input is not found in hospital_path:
                Print ("This code does not exist, please try again")

            Else:
                Loop until True:
                    Prompt user ("Which detail do you want to edit? [Name])
                    Set hospital_path to read +
                    If user input == "Name":
                        Prompt user ("Please enter the new name")
                        Edit hospital to new hospital name #By hospital Code
                        Break

                    Else:
                        Print ("Please enter the correct details.")

                End If

            End Loop

        End If

    End Loop

Else:
    Print("Please create hospital file before distribute inventory")
    Call Function hospital_file(hospital_path)

End If

Else if user inputs 5: #Track Inventory Item
    Loop until True:
        Prompt user ("Do you want to track available quantity of all items or track all items which has less than 25 stock? [1 or 2]")
        If user inputs 1:
            Print ppe.txt in ascending order by item code
            Break

        Else if user inputs 2:
            Check ppe.txt for items where item_quantity < 25
            Print items where item_quantity < 25
            Break

        Else:
            Print ("The input is invalid. Please try again.")

    End If

End Loop

```

```

Else if user inputs 6: #Search Distribution List
    Loop until True:
        Prompt user ("Please enter the Item Code")
        If user input is not found in ppe_path:
            Print ("This input is invalid. Please try again.")

        Else:
            If user input is not found in distribution_path:
                Print(user input, "does not distribute to any hospital yet.")

            Else:
                Open distribution_path file in read mode
                Hospital_quantity = {} #Create Dictionary to sum up the quantity for same hospital
                For each line in distribution_path:
                    Split line into parts by ","

                    #Sum up Quantity for same hospital
                    If hospital_code in hospital_quantity:
                        hospital_quantity[hospital_code] += quantity
                    Else:
                        hospital_quantity[hospital_code] = quantity

                End If

            End For Loop

            For hospital_code, quantity in hospital_quantity.items():
                Print (user input, "has been distributed", quantity, "to", hospital_code)
                Break

            End For Loop

        End If

    End Loop

Else if user inputs 7: #Print Report
    Loop until True:
        Prompt user ("Would you like to print the list of suppliers with their PPE equipments supplied or the list of hospitals with
        quantity of distribution items or overall transaction report for a selected month?
        \n "1" to Print the List of Suppliers with their PPE equipments supplied
        \n "2" to Print the List of Hospitals with quantity of distribution items
        \n "3" to Print the Overall Transaction Report for a selected month")

        If user inputs 1:
            Open supplier_path in read mode
            suppliers = {} #Create dictionary
            For each line in supplier_path:
                Split line into parts by ","
                Extract Supplier_code, supplier_name, and add into suppliers dictionary

            End For Loop
            Close supplier_path file

            Open ppe_path in read mode
            items = {} #Create dictionary
            For each line in ppe_path:
                Split line into parts by ","
                Extract item_code, item_name, item_quantity, supplier_code
                If supplier_code in suppliers:
                    If supplier_code not in items:
                        Initialize items[supplier_code] as an empty list
                    End If
                    Append {item_code, item_name, item_quantity} to items[supplier_code]
                End If

            End For Loop
            Close ppe_path file
            For supplier_code in items:
                Print("Supplier", suppliers[supplier_code])
                For item_info in items[supplier_code]:
                    Print("Item:", item_info[0], item_info[1], "Quantity", item_info[2])
                    #Expected output
                    # Supplier: supplier_code
                    # Item: item_code, item_name, Quantity: [number]

            End For Loop

        Else if user inputs 2: #List of Hospitals with distribution
            Call Function distribution_hospital()

        Else if user inputs 3: #Overall Transaction Report

    Loop until True:
        Open distribution_path file in read mode

```

```
        Open ppe_path file in read mode
        Print ("Please input the month which you would like to retrieve data from:")

        If month in distribution.txt and ppe.txt
            Print ("Transaction Report for items distributed in", month)
            Print item code, item quantity, hospital code where date starts with month inputted
            Print ("Transaction Report for items received from suppliers in", month)
            Print item code, amount added where date starts with month inputted
            Break
        Else
            Print ("The month entered is invalid")
        End If

        Close distribution_path file
        Close ppe_path file
    End Loop

Else:
    Print("Please enter correct input")
End If

End Loop

Else if user input 8:
    Print("Thank you for using this system. You have successfully logged out")
    Return #Logout
Else:
    Print("Please enter the correct input")
End If

End Loop

END
```

#Main Code

START

Function main():

 Set user_count = 0 #max 4

 Users = {} #create dictionary

 Loop until True:

 Prompt user “Would you like to sign up or log in?”

 Prompt “1” for user sign up

 Prompt “2” for user log in

 If user inputs 1:

 Call function sign_up

 Call function home

 Else if user inputs 2:

 If Call function login(users, attempt) returns True:

 Call function home

 Else:

 Print(“Please enter correct input”)

 Break

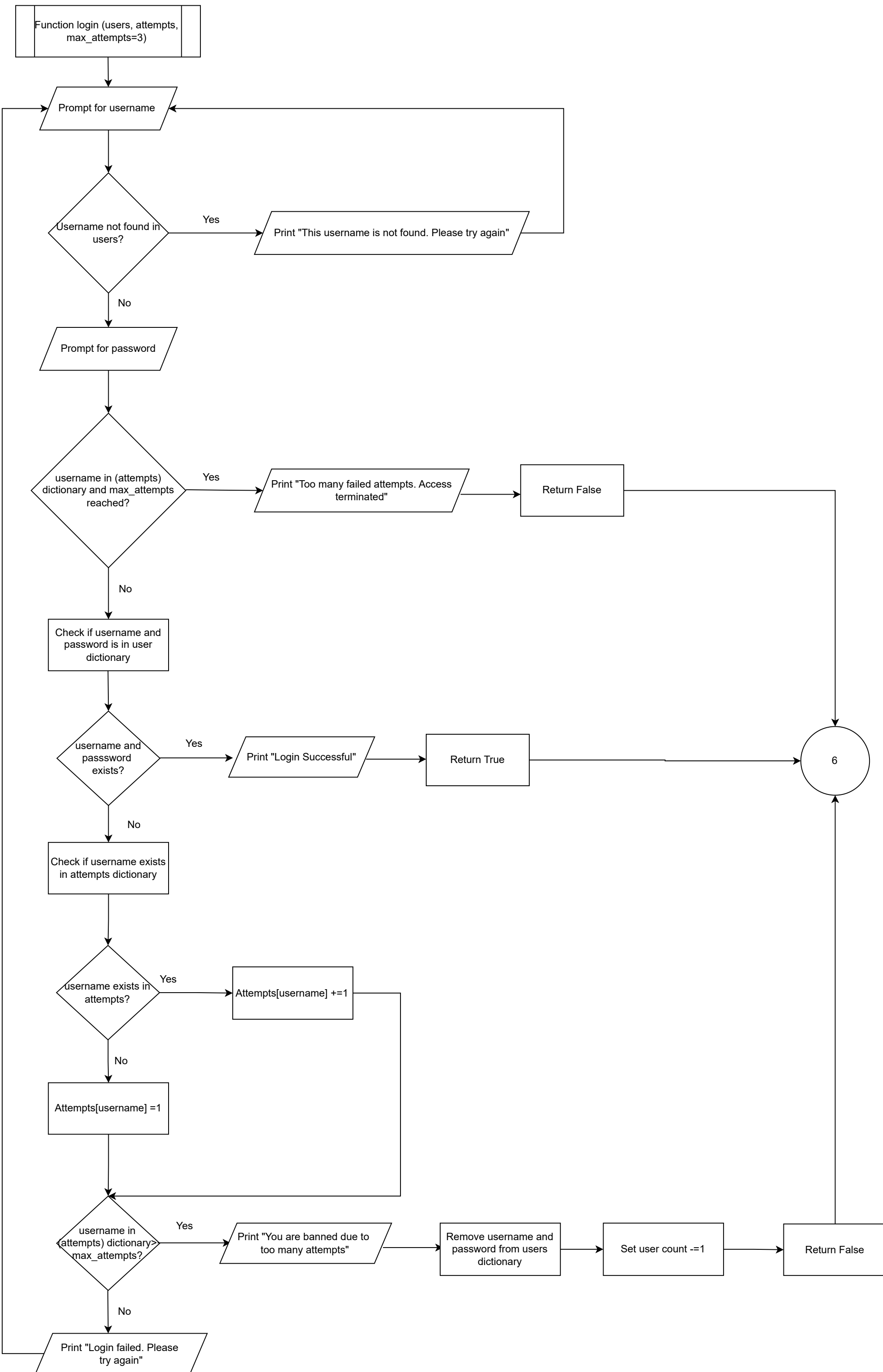
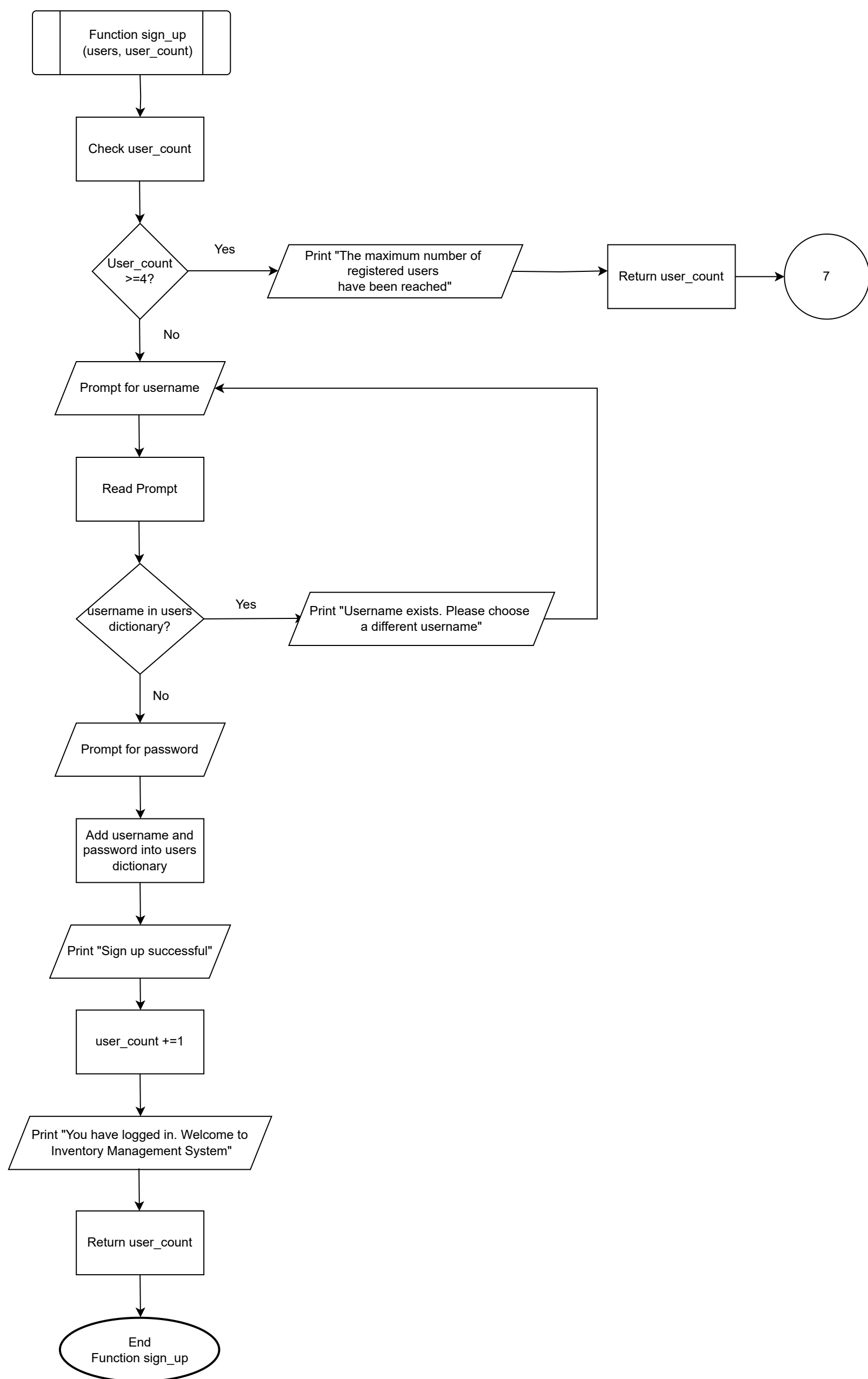
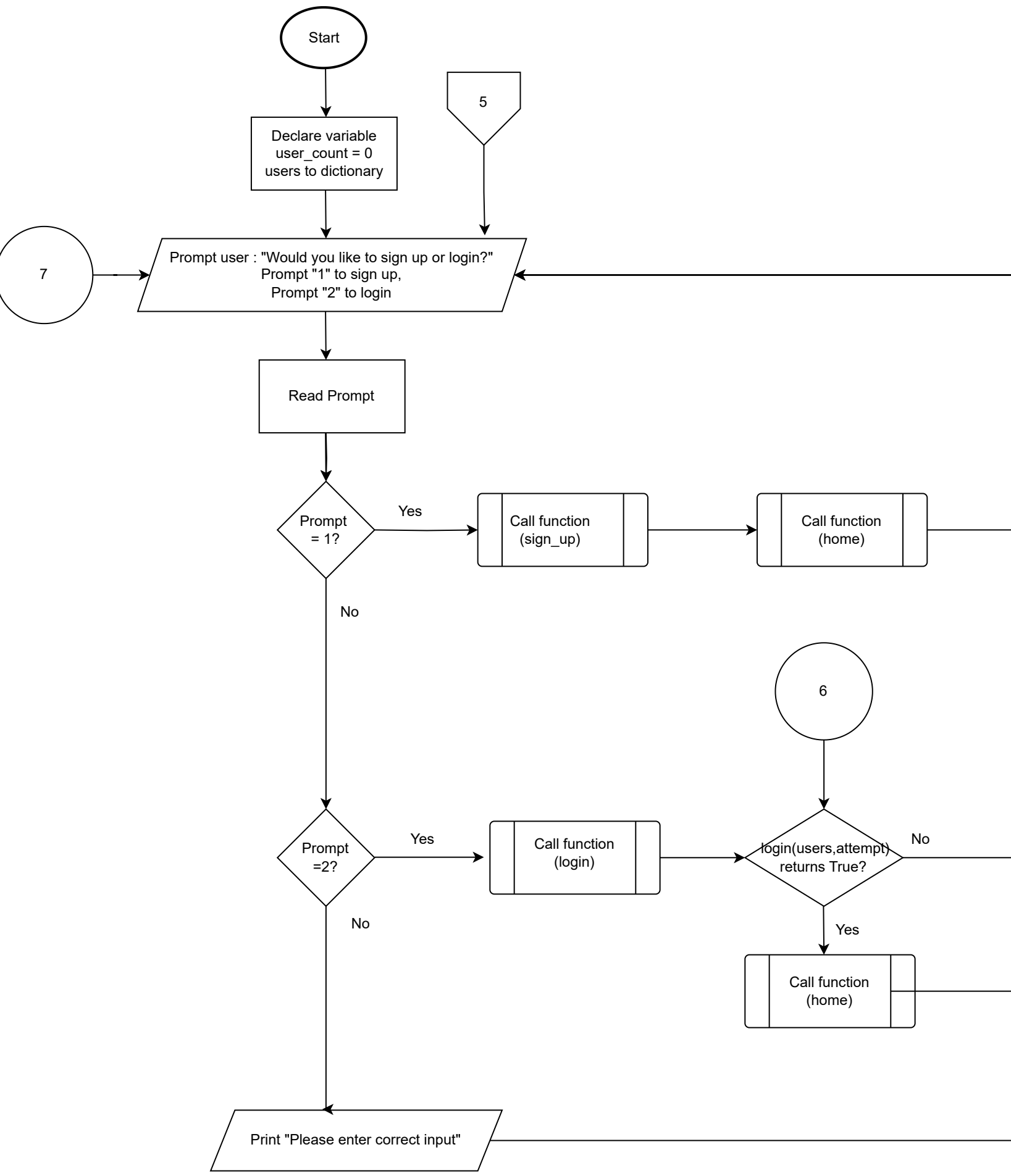
 End If

 End Loop

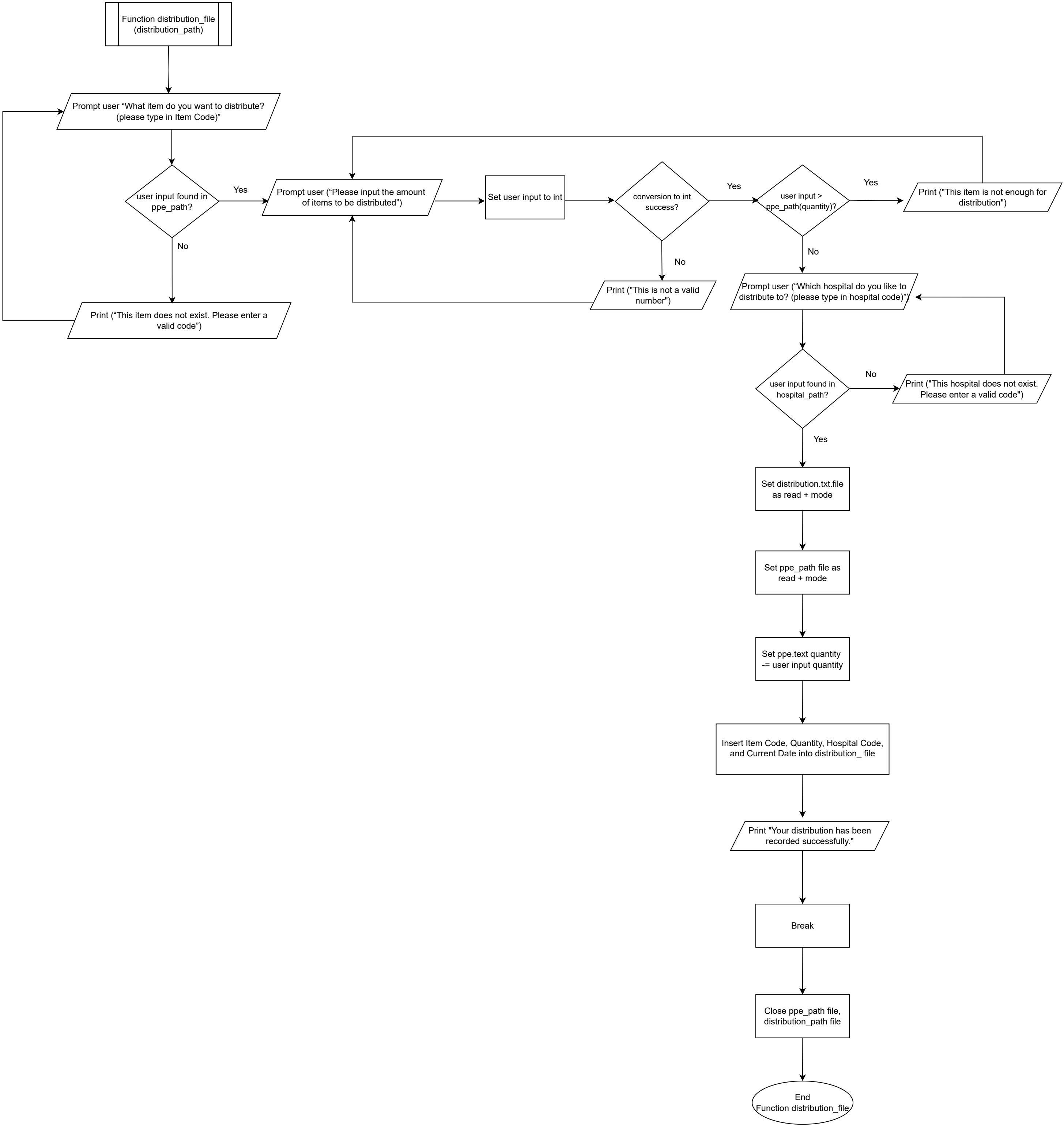
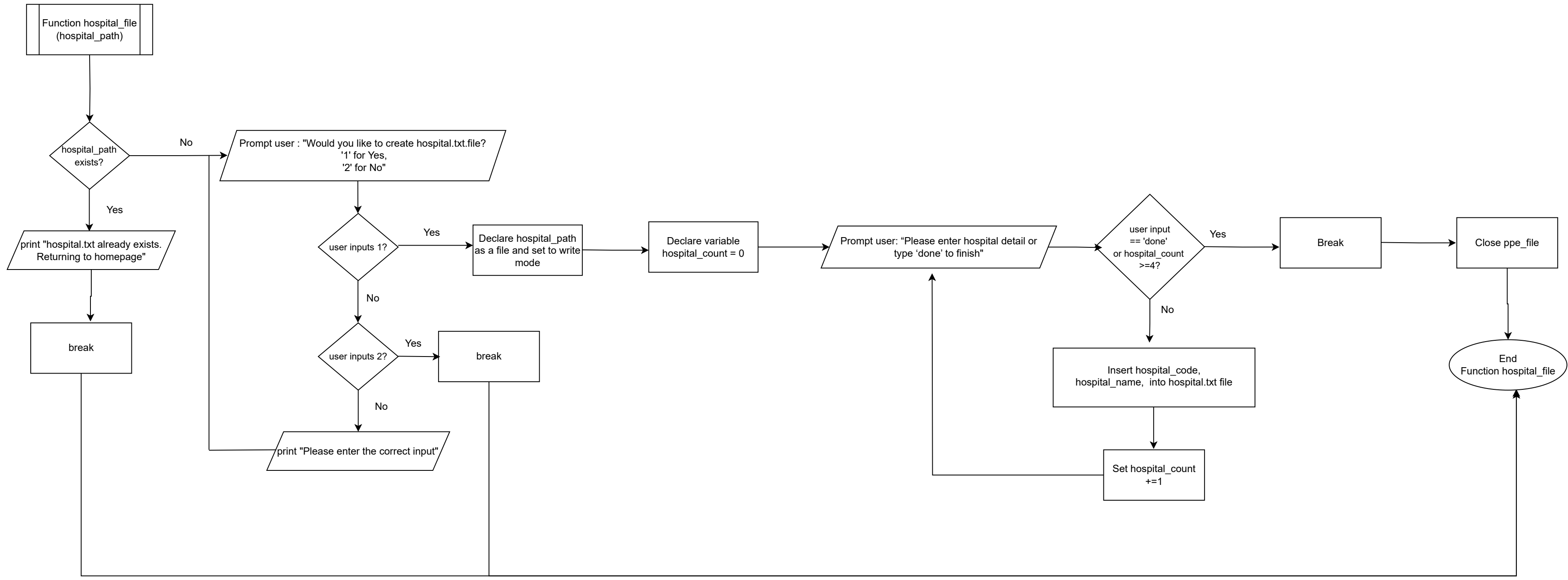
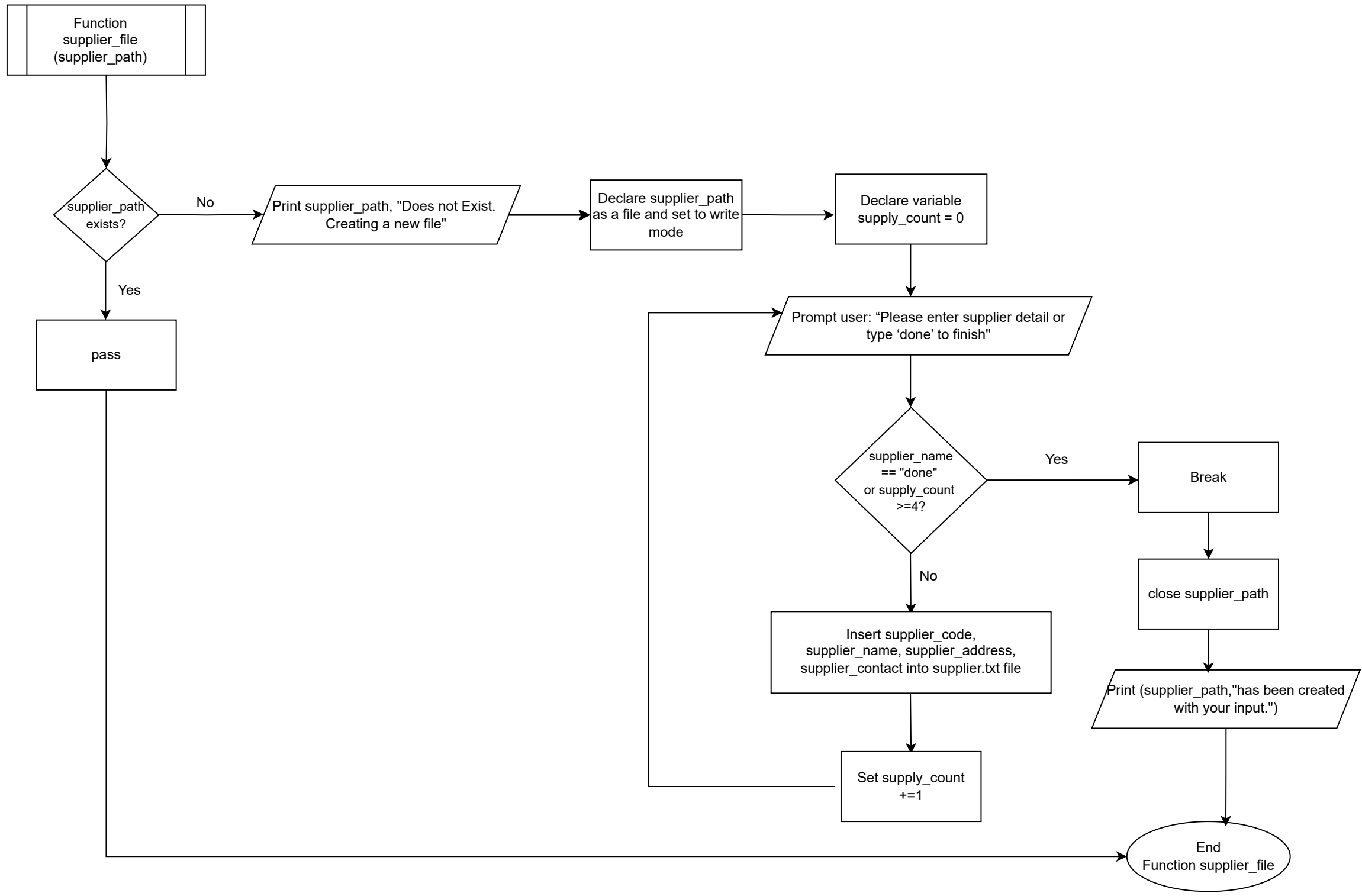
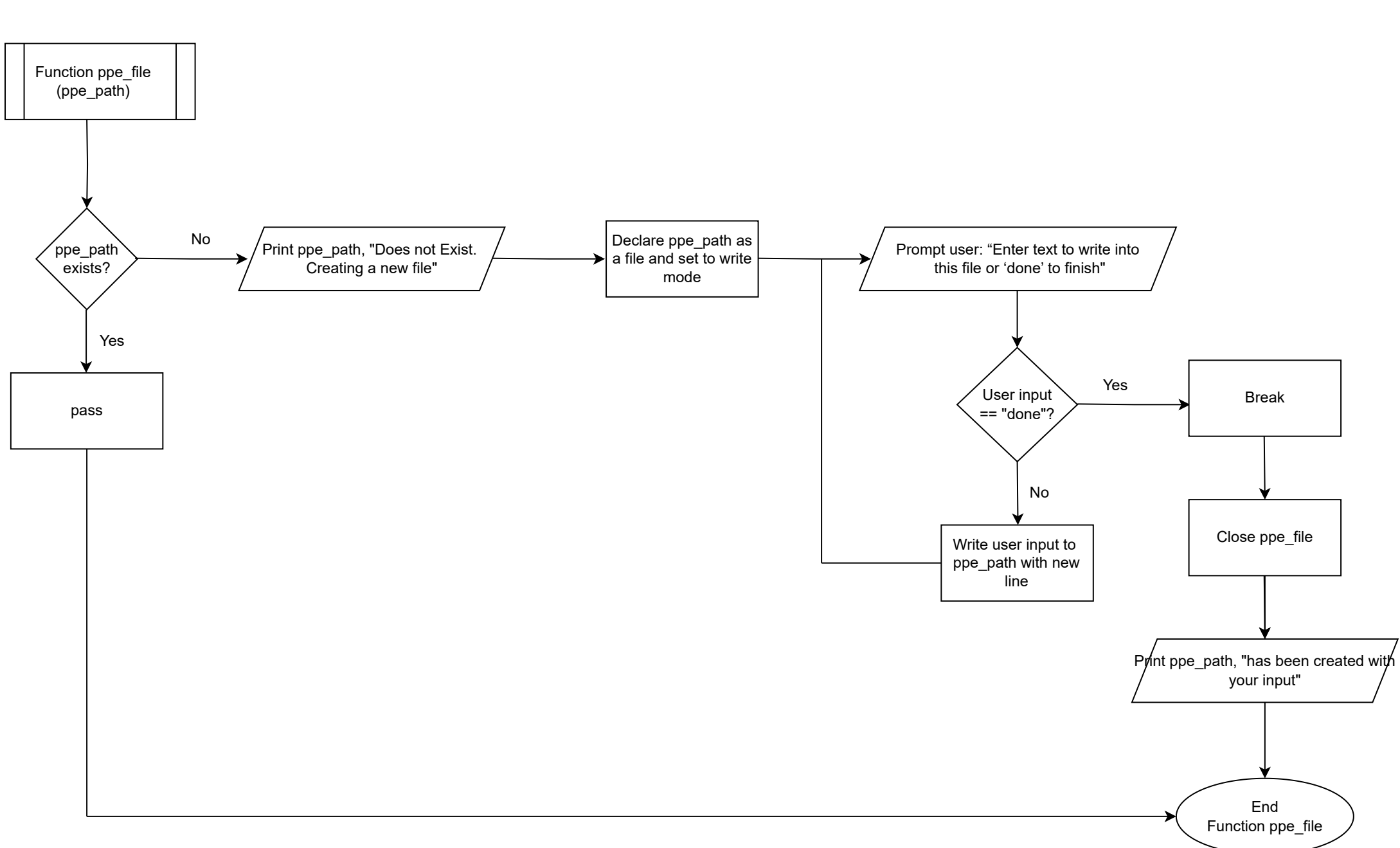
END

4.2 Flowchart

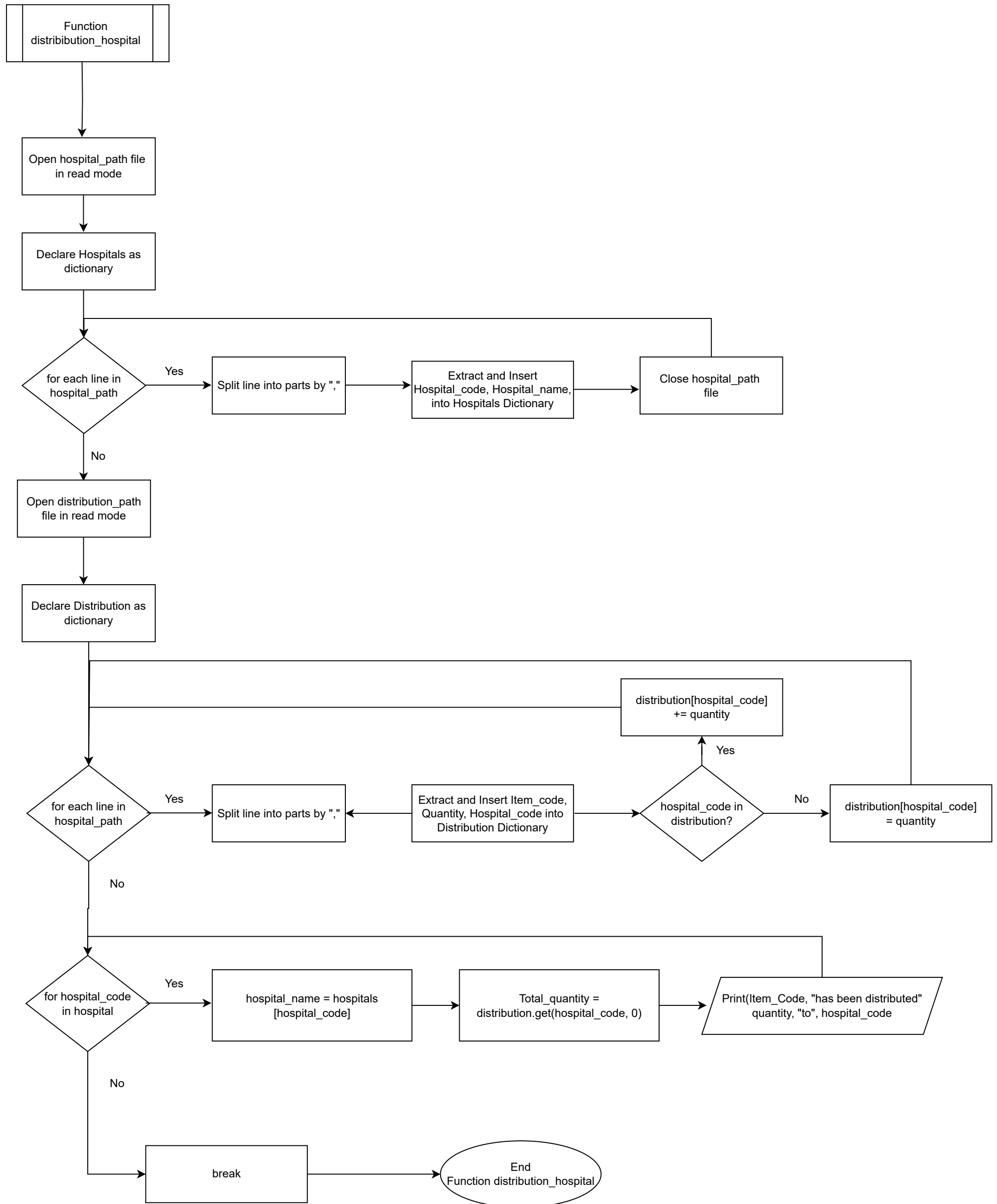
Main +login/signup



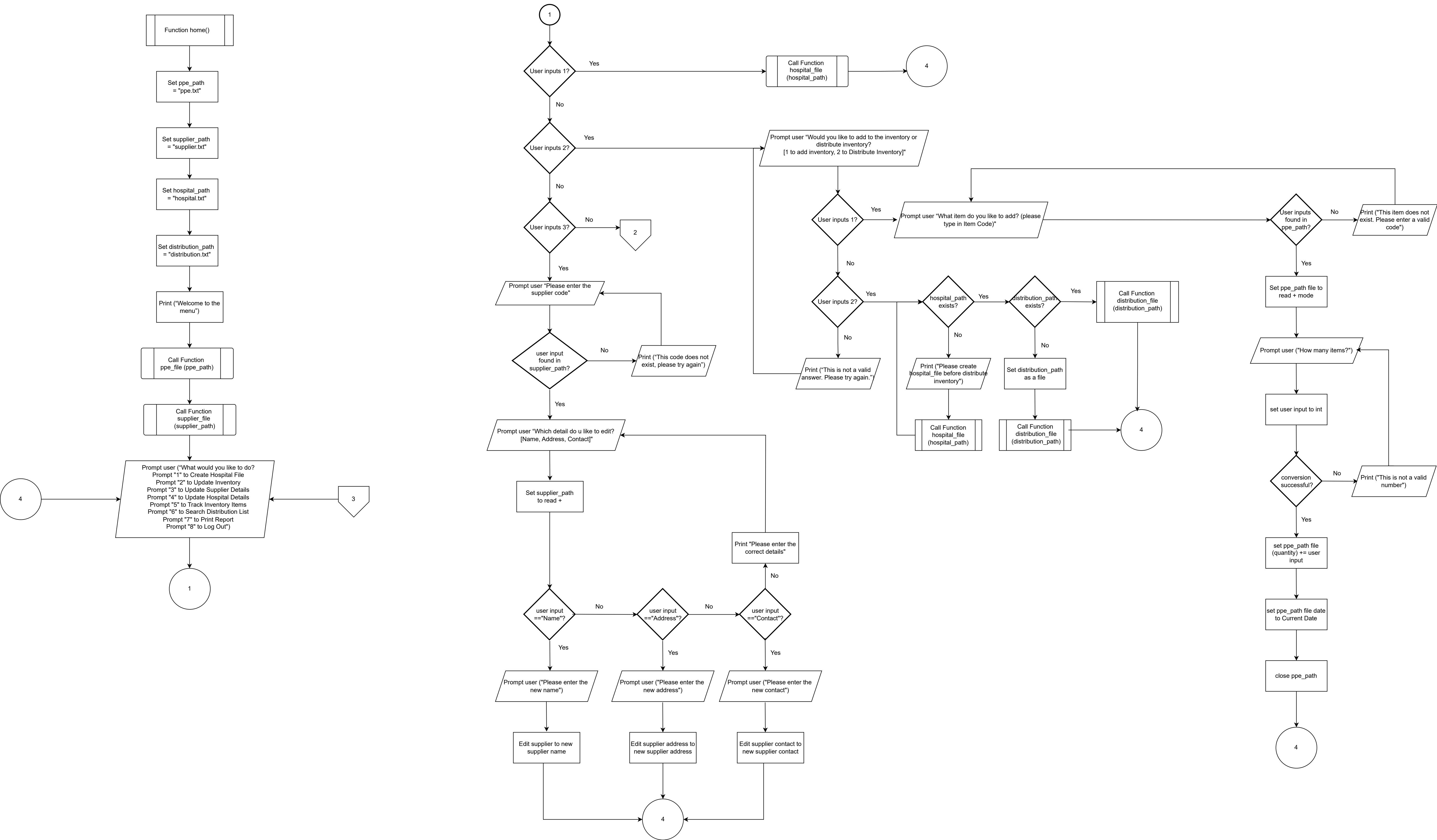
Functions_txt.file



Function- Distribution_hosp



Function_Home



Function_Home_2

